RUIHAN QIN

Selected work.

Turning questions into working systems.

I work across mathematical modeling, frontend and backend development, and AI agents design to explore questions and build tools people can use.

Websites 02-03
Research & Modeling 04-07
Awards 08

Explore the interactive portfolio

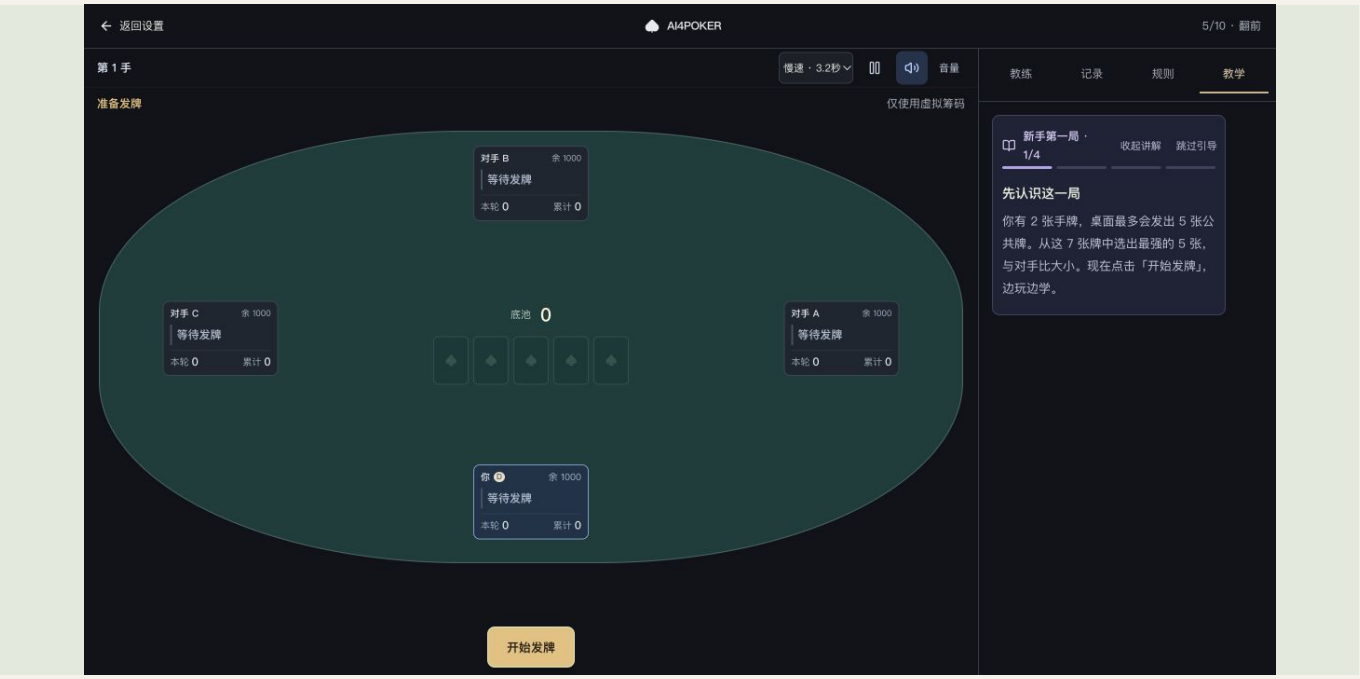
Live products, project details, original reports and award certificates.

gambit025.github.io/portfolio

AI4POKER

Learn the game. Understand the decision.

A browser-based Texas Hold'em learning environment that combines a local game engine, guided practice and contextual coaching.



AI4POKER: the practice table and contextual teaching panel.

A complete practice environment

Configurable tables, betting rounds, hand history and settlement bring the rules into a working game.

Computation off the main thread

A Web Worker estimates hand equity while the interface stays focused on the table and the player's next action.

Technology stack

- React & TypeScript
- Vite & Tailwind CSS
- Web Workers
- Web Audio API
- Cloudflare Pages

Experience AI4POKER

One-Stop Resume

From an existing résumé to a polished PDF.

A full-stack document tool that extracts résumé content, structures it with AI and renders a new, template-based PDF.



One-Stop Resume: an upload workflow alongside the document preview.

A complete document pipeline

PyMuPDF extracts text, Groq structures the content, and deterministic HTML/CSS templates are rendered to PDF with Chromium.

Preview the actual output

Page-by-page previews and zoom controls let users inspect the generated document before downloading it.

Technology stack

- HTML, CSS & JavaScript
- Python & Flask
- PyMuPDF
- Groq API
- Playwright / Chromium
- Docker & Render

Experience One-Stop Resume

Multimodal AI & virtual-human simulation

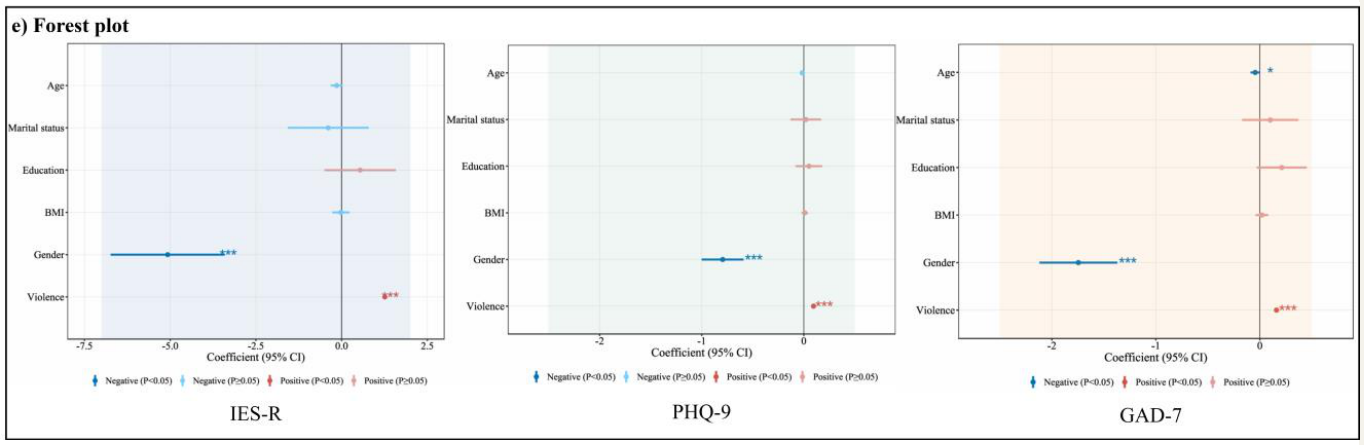
My contribution

Co-author: research direction, programming, data analysis and visualization.

The study connects five-dimensional video-intensity scoring with a cohort of 1,154 generated virtual humans, then analyzes responses to exposure and informational interventions.

Model stack & analysis

PandaGPT · ImageBind & Vicuna · CO-STAR prompting
Linear regression · Mixed-effects models



Manuscript Figure 3, detail: regression results for the generated cohort.

Research contribution

I helped define the direction of the study, implemented parts of the computational workflow, analyzed the data and produced research visualizations.

Research design

The pipeline combines multimodal signal processing, virtual-human simulation and longitudinal statistical analysis.

Manuscript prepared for Journal of Affective Disorders. Summary and selected figures only.

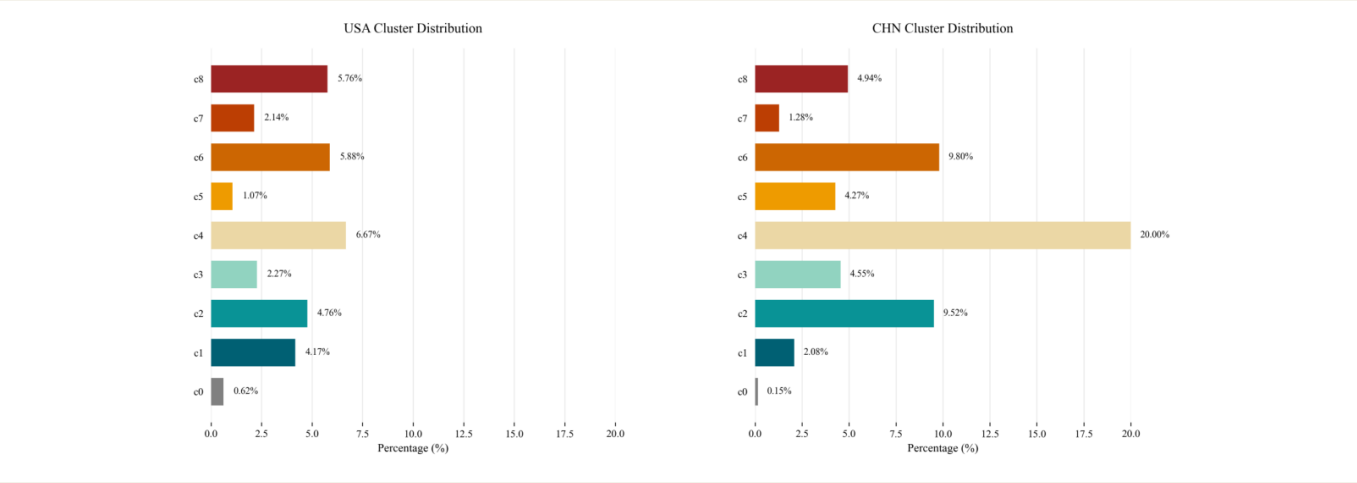
M3POC

Forecasting medals. Examining the drivers.

A competition project that combines athlete-level features, medal forecasting and analysis of coaching effects.

My contribution

Team project: mathematical modeling and programming.



Original report, Figures 3-4: sport-cluster distributions used in the forecasting framework.

Different models for different questions

Clustering grades sports, forecasting models estimate medal counts, and logistic regression examines first-medal breakthroughs.

Performance and context together

The framework considers athlete strength, participation, new entrants and host-country effects.

Modeling methods

- K-means clustering
- Random forest
- ARIMA
- Logistic regression
- Difference-in-differences

Honorable Mention in the 2025 Mathematical Contest in Modeling.

Production decisions under uncertainty

Better sampling. Better production decisions.

A computational framework for choosing inspection and disassembly strategies in electronics manufacturing.

My contribution

Team project: mathematical modeling and programming.



Original report: profit comparison across inspection and disassembly strategies.

Adaptive sampling

Sequential and hypothesis testing support accept-or-reject decisions as inspection evidence accumulates.

Simulation of decision alternatives

Monte Carlo experiments compare inspection and disassembly strategies under the supplied cost and defect assumptions.

Tools & methods

- Python & C++
- NumPy & Matplotlib
- Sequential testing
- Monte Carlo simulation
- Genetic algorithms

National Second Prize in the 2024 China Undergraduate Mathematical Contest in Modeling.

QuantDIA

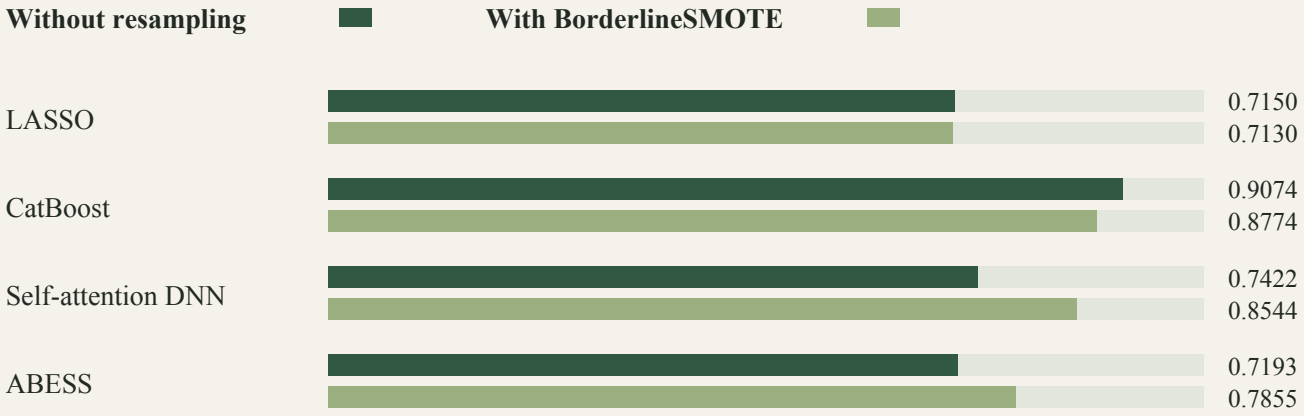
Prediction with a molecular explanation.

An interpretable machine-learning framework for studying drug-induced autoimmunity from molecular descriptors.

My contribution

Team project: mathematical modeling and programming.

AUC by model and sampling condition



Source: competition report. Bar lengths use the full 0-1 AUC scale.

Four complementary models

LASSO, ABESS, CatBoost and a self-attention neural network offer different approaches to feature selection and prediction.

Explicit treatment of class imbalance

The report compares models with and without BorderlineSMOTE on a dataset of 597 drugs and 196 descriptors.

Technology stack

- Python
- scikit-learn
- CatBoost
- PyTorch
- ABESS
- Matplotlib

Regional Second Prize in the 2025 National College Student Statistical Modeling Competition.

Awards

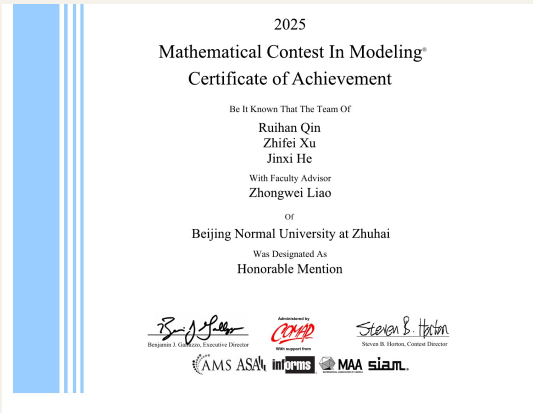


2024

China Undergraduate Mathematical Contest in Modeling

National Second Prize

View original certificate



2025

Mathematical Contest in Modeling

Honorable Mention

View original certificate



2025

National College Student Statistical Modeling Competition

Regional Second Prize

View original certificate

View the full interactive portfolio

gambit025.github.io/portfolio